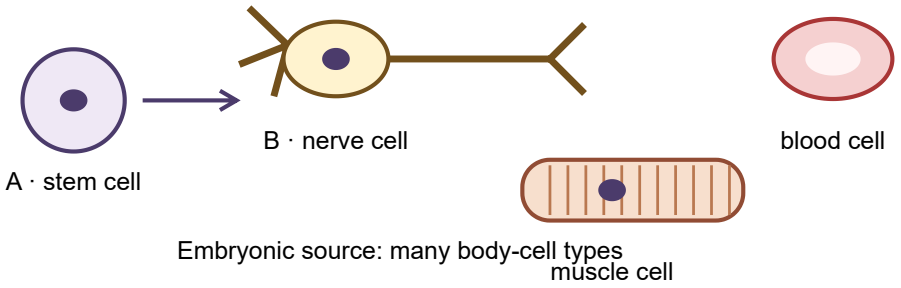
Reference A animal cells



An embryonic stem cell can give rise to many specialised body-cell types.

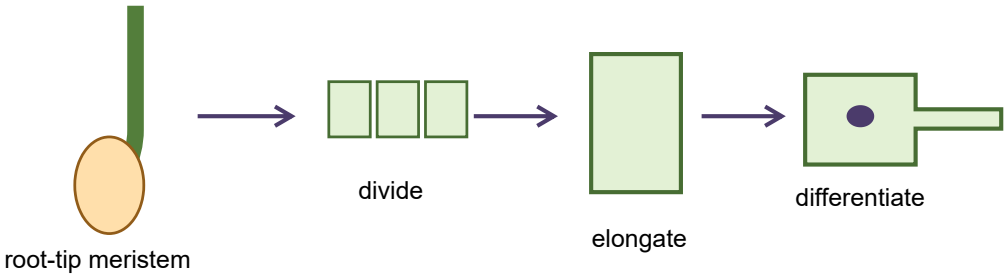
Reference only. Do not write on this sheet. Trace a feature, then explain its function.

Cell or source	Feature or range	Function or limit
Nerve cell	Long extension	Carries signals over a distance
Muscle cell	Structures that contract	Contraction produces force
Embryonic stem cell	Can give rise to many body-cell types	Not a promise of a successful treatment
Adult blood-forming stem cell	More limited range of blood-cell types	Makes replacement blood cells

The arrows show possible lineages. The drawing does not show every cell stage. Cells do not choose their own jobs. Specialised cells can share inherited DNA but use it differently.

Made by Matt

Reference B root growth



Root-tip meristem, division, elongation and differentiation in a simplified route.

Reference only. Do not write on this sheet.

Process or region	What it adds	Useful limit
Meristem	A region where cells divide	Not every root cell remains a meristem cell
Division	More cells	Number alone does not prove a specialised role
Elongation	Longer cells	Not the name for becoming specialised
Differentiation	Features suited to a job	Different cells can be at different stages

A root hair extension provides a large surface for absorption. The route is simplified. Use it to compare processes, not as a claim that every cell follows an identical sequence.

Made by Matt

Arrival and exit Supported

Use the labelled picture on Reference A for question 1. Respond in your usual mode.

1. In the picture, which cell is labelled stem cell: A or B?

2. Cell division makes more cells or fewer cells?

3. What holds chromosomes in a eukaryotic animal cell: nucleus or cell membrane?

4. A root tip contains a meristem. Point to root tip or leaf middle as a place that makes new plant cells.

Exit ticket

1. Point to division or differentiation for a cell gaining a job.

2. Choose more cells or longer cells for the direct result of division.

Pupil voice

What helped you most today, or what would you change next time?

Point, speak, sign, draw, write or choose. You may pass.

verbal · written · drawn · signed · symbol / choice-selected · demonstrated / re-attempt · adult-scribed. You may pass.

Made by Matt

You do Supported

Use your book or this sheet. One response per task. Use Reference A and B.

1. Finish: Division makes more ___. Differentiation gives a specialised ___. [2] Staff I ☐ S ☐ C ☐

Use cells and job. Point to the two changes in Reference A.

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2. Use Reference A. Choose nerve or muscle for a cell with a long extension. Say how that feature helps. [3] Staff I ☐ S ☐ C ☐

A nerve cell carries signals. Link the long extension to carrying a signal over a distance.

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3. Choose the possible route: an adult blood-forming stem cell makes blood cells or every kind of body cell. Give a reason. [3] Staff I ☐ S ☐ C ☐

The source has a limited range of cell types. Use the blood-cell branch.

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4. Put the plant cards in order: differentiate, divide, elongate. Name the region where new cells are made. [4] Staff I ☐ S ☐ C ☐

Start at the root tip. More cells come before longer cells in this simplified route.

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5. Compare the skin and root models. Finish: Both make more cells by ___. The root cells also ___. A cell gains a job by ___. [3] Staff I ☐ S ☐ C ☐

Use division, elongate and differentiation. Use the order as a teaching model, not a rule for every cell.

Feedback and response · Supported

Feedback codes

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt